EDUGENIE

Google Gemini Powered Learning Assistant

Complete Project Program

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1. requirements.txt

Copy this code into the file with the same name.

```
fastapi
uvicorn
google-generativeai
jinja2
python-multipart
```

2. main.py

Copy this code into the file with the same name.

```
from fastapi import FastAPI, Request, Form
from fastapi.templating import Jinja2Templates
from fastapi.staticfiles import StaticFiles

from qna import ask_question
from explanation_module import explain_topic
from quiz_module import generate_quiz
from summary_module import summarize_text
from learning_path import get_learning_path

app = FastAPI(title="EduGenie")

app.mount("/static", StaticFiles(directory="static"), name="static")
templates = Jinja2Templates(directory="templates")


@app.get("/")
def home(request: Request):
    return templates.TemplateResponse(
        "index.html",
        {"request": request, "result": ""}
    )


@app.post("/process")
def process(
    request: Request,
    task: str = Form(...),
    text: str = Form(...)
):
    if task == "qa":
        result = ask_question(text)
    elif task == "explain":
        result = explain_topic(text)
    elif task == "quiz":
        result = generate_quiz(text)
    elif task == "summary":
        result = summarize_text(text)
    elif task == "learning":
        result = get_learning_path(text)
    else:
        result = "Invalid task selected."

    return templates.TemplateResponse(
        "index.html",
        {"request": request, "result": result}
    )
```

3. qna.py

Copy this code into the file with the same name.

```
import google.generativeai as genai

API_KEY = "YOUR_GEMINI_API_KEY"
genai.configure(api_key=API_KEY)
model = genai.GenerativeModel("gemini-1.5-flash")


def ask_question(question):
    prompt = f"""
    You are EduGenie, an educational AI assistant.
    Answer the following student question clearly and accurately.

    Question:
    {question}

    Give a concise and easy-to-understand answer.
    """
    try:
        response = model.generate_content(prompt)
        return response.text
    except Exception as e:
        return "Error: " + str(e)
```

4. explanation_module.py

Copy this code into the file with the same name.

```
import google.generativeai as genai

API_KEY = "YOUR_GEMINI_API_KEY"
genai.configure(api_key=API_KEY)
model = genai.GenerativeModel("gemini-1.5-flash")


def explain_topic(topic):
    prompt = f"""
    Explain the following topic to a beginner.

    Topic:
    {topic}

    Requirements:
    - Use simple English.
    - Give a clear definition.
    - Explain the important points.
    - Give a simple example.
    """
    try:
        response = model.generate_content(prompt)
        return response.text
    except Exception as e:
        return "Error: " + str(e)
```

5. quiz_module.py

Copy this code into the file with the same name.

```
import google.generativeai as genai

API_KEY = "YOUR_GEMINI_API_KEY"
genai.configure(api_key=API_KEY)
model = genai.GenerativeModel("gemini-1.5-flash")


def generate_quiz(topic):
    prompt = f"""
    Create a quiz for the following topic:

    Topic:
    {topic}

    Generate exactly 3 multiple-choice questions.

    Each question must contain:
    A, B, C and D options.

    Also provide the correct answer.

    Format:

    Question 1:
    A)
    B)
    C)
    D)
    Answer:

    Question 2:
    A)
    B)
    C)
    D)
    Answer:

    Question 3:
    A)
    B)
    C)
    D)
    Answer:
    """
    try:
        response = model.generate_content(prompt)
        return response.text
    except Exception as e:
        return "Error: " + str(e)
```

6. summary_module.py

Copy this code into the file with the same name.

```
import google.generativeai as genai

API_KEY = "YOUR_GEMINI_API_KEY"
genai.configure(api_key=API_KEY)
model = genai.GenerativeModel("gemini-1.5-flash")


def summarize_text(text):
    prompt = f"""
    Summarize the following educational text.

    Text:
    {text}

    Requirements:
    - Keep the important information.
    - Remove unnecessary details.
    - Use simple English.
    - Present the result in short paragraphs or bullet points.
    """
    try:
        response = model.generate_content(prompt)
        return response.text
    except Exception as e:
        return "Error: " + str(e)
```

7. learning_path.py

Copy this code into the file with the same name.

```
import google.generativeai as genai

API_KEY = "YOUR_GEMINI_API_KEY"
genai.configure(api_key=API_KEY)
model = genai.GenerativeModel("gemini-1.5-flash")


def get_learning_path(topic):
    prompt = f"""
    Create a structured learning path for:

    Topic:
    {topic}

    Include:
    1. Beginner topics
    2. Intermediate topics
    3. Advanced topics
    4. Suggested learning order
    5. Practice activities
    6. Useful learning resources

    Explain everything in simple English.
    """
    try:
        response = model.generate_content(prompt)
        return response.text
    except Exception as e:
        return "Error: " + str(e)
```

8. templates/index.html

Copy this code into the file with the same name.

```
<!DOCTYPE html>
<html>
<head>
    <title>EduGenie - AI Learning Assistant</title>
    <link rel="stylesheet"
          href="{{ url_for('static', path='style.css') }}">
</head>

<body>
<div class="container">
    <h1>EduGenie</h1>
    <h2>Google Gemini Powered Learning Assistant</h2>

    <p class="intro">
        Learn smarter with AI-powered explanations,
        answers, quizzes, summaries and learning paths.
    </p>

    <form action="/process" method="post">
        <label>Select Task</label>

        <select name="task">
            <option value="qa">Ask a Question</option>
            <option value="explain">Explain a Topic</option>
            <option value="quiz">Generate Quiz</option>
            <option value="summary">Summarize Text</option>
            <option value="learning">Learning Path</option>
        </select>

        <label>Enter your topic or question</label>

        <textarea name="text"
                  placeholder="Enter your question or topic here..."
                  required></textarea>

        <button type="submit">Generate Result</button>
    </form>

    {% if result %}
    <div class="result">
        <h3>EduGenie Result</h3>
        <pre>{{ result }}</pre>
    </div>
    {% endif %}
</div>
</body>
</html>
```

9. static/style.css

Copy this code into the file with the same name.

```
body {
    font-family: Arial, sans-serif;
    background: linear-gradient(135deg, #667eea, #764ba2);
    margin: 0;
    padding: 40px;
}

.container {
    max-width: 850px;
    margin: auto;
    background: white;
    padding: 35px;
    border-radius: 20px;
    box-shadow: 0px 10px 30px rgba(0,0,0,0.2);
}

h1 {
    text-align: center;
    color: #5a3ec8;
    font-size: 42px;
    margin-bottom: 5px;
}

h2 {
    text-align: center;
    color: #444;
}

.intro {
    text-align: center;
    color: #666;
    margin-bottom: 30px;
}

label {
    display: block;
    margin-top: 20px;
    margin-bottom: 8px;
    font-weight: bold;
}

select, textarea {
    width: 100%;
    padding: 12px;
    border: 1px solid #ccc;
    border-radius: 10px;
    box-sizing: border-box;
    font-size: 16px;
}

textarea {
    height: 150px;
    resize: vertical;
}

button {
    width: 100%;
    margin-top: 20px;
    padding: 14px;
    border: none;
    border-radius: 10px;
    background: #5a3ec8;
    color: white;
    font-size: 18px;
    cursor: pointer;
}

button:hover {
    background: #452aa5;
}

.result {
    margin-top: 30px;
    padding: 20px;
    background: #f4f1ff;
    border-radius: 12px;
    border-left: 5px solid #5a3ec8;
}
```

```
.result h3 {
    color: #5a3ec8;
}

pre {
    white-space: pre-wrap;
    font-family: Arial, sans-serif;
    line-height: 1.6;
}
```

HOW TO RUN THE PROJECT

1. Create the folder structure shown in the report.
2. Install Python 3.10 or later.
3. Open terminal inside the EduGenie folder.
4. Run: pip install -r requirements.txt
5. Replace YOUR_GEMINI_API_KEY with your Gemini API key in the five Python modules.
6. Run: uvicorn main:app --reload
7. Open: http://127.0.0.1:8000

Important: Keep the API key private and do not publish it in GitHub or screenshots.